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Addressing Mobility Overestimation in Short-Channel IGZO TFTs Using the Gated Van der Pauw Method

Woo-Seok Lee, Jaeho Lee, Amarja Katware, Noh-Hwal Park, Jiyoung Kim,* Rino Choi,* and Jeong-Hwan Lee*

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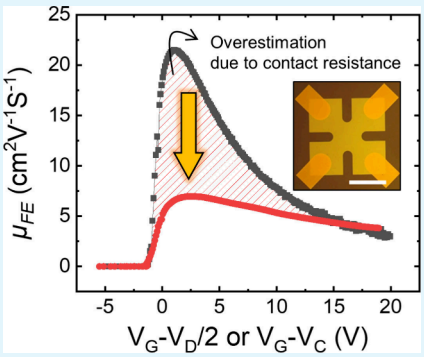
ABSTRACT: The determination of field-effect mobility in indium gallium zinc oxide (IGZO) thin-film transistors (TFTs) is pivotal for advancing high-performance electronic devices. However, accurately evaluating the field-effect mobility in IGZO-TFTs with short channel lengths of less than a few micrometers poses a formidable challenge. This difficulty arises due to the increasing significance of contact resistance in the total resistance as the channel length decreases. Consequently, increasing contact resistance can lead to inaccurate sheet resistance values, resulting in overestimating critical parameters, such as mobility and contact resistance, during device evaluations. Therefore, a precise method for extracting sheet resistance, an intrinsic channel characteristic, is imperative for short-channel IGZO TFTs. Herein, a gated van der Pauw (gVDP) method was employed to extract sheet resistance precisely and independently of contact resistance. This contact-resistance-independent approach mitigates the risk of overestimating charge behaviors in IGZO TFTs. Furthermore, it facilitates precise evaluations of intrinsic channel mobility, enabling accurate predictions of the maximum performance of devices with short channels.



KEYWORDS: *indium−gallium−zinc-oxide thin-film transistors, contact resistance, gated van der Pauw, sheet resistance, field-effect mobility*

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1. INTRODUCTION

Amorphous indium−gallium−zinc oxide (a-IGZO) thin-film transistors (TFTs) are promising channel materials owing to their high field-effect mobility, exceptional uniformity, low off-current with a high on−off ratio, strong thermal resistance, and small subthreshold swing. Their compatibility with micro/nanosemiconductor processes at low temperatures (<300 °C) makes them advantageous for practical applications.[1−6] Consequently, they are attracting increasing attention for use in a wide spectrum of devices, ranging from display backplanes to next-generation memory, logic, and neuromorphic applications.[7−13] The increasing demand for enhanced electrical performance, encompassing mobility and low power consumption, has highlighted the necessity to downscale IGZO TFTs.[14−18] In transistors with a long channel length, the contact resistance (R_C) is minimal compared to the channel resistance (R_{Ch}), exerting a minimal influence on the total resistance (R_T) and field-effect mobility (μ_{FE}). On the other hand, the precise extraction of the μ_{FE} is challenging when devices are scaled down to a channel length of less than a few micrometers.[19−26]

The main challenge in accurately extracting mobility for short-channel oxide TFTs is that mobility estimation relies on MOSFET models, which assume negligible contact resistance. While the MOSFET model is a powerful tool for analyzing silicon-based transistors, the method could result in errors when applied to analyzing transistors with significant R_C, such as oxide TFTs. This discrepancy arises because actual devices exhibit a nonlinear I_D−V_G relationship due to the inclusion of R_C, contrasting with the ideal linear I_D−V_G relationship assumed by the model.[27,28] Liu et al. claimed that this nonideal behavior is due to a potential drop caused by the contact barrier in the TFT. The overestimated peak mobility is related to the presence of R_C in TFTs.[29−31] As a result, as the channel length is shortened, the model becomes less representative of the actual device, leading to an overestimation of mobility. To address this issue, Bittle et al. employed impedance analysis to interpret the nonideal behavior in I_D−V_G.[32] This approach corrects the errors in the model caused by R_C behavior at low V_G values by utilizing the actual carrier charge. Additionally, Pei et al. reported that changes in the W/L ratio due to the fringe effect could lead to overestimating mobility.[33] Another study documented issues related to carrier injection in transistors with nonohmic

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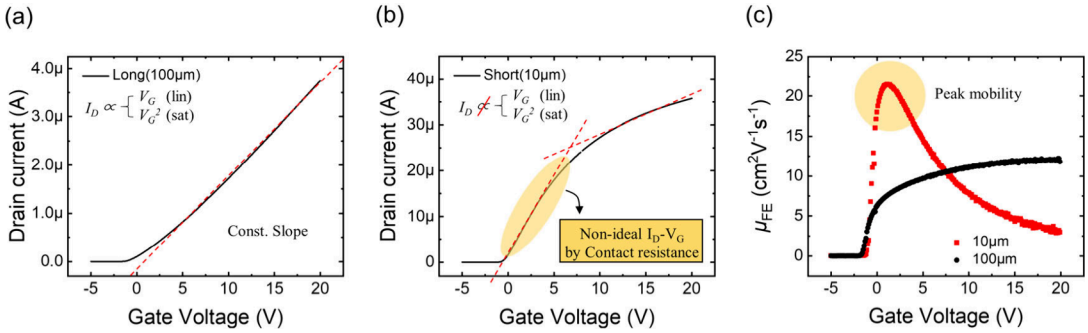

Figure 1. I_D−V_G characteristics for (a) a long-channel a-IGZO TFT with a length of 100 μm and (b) a short-channel TFT with a length of 10 μm in the linear regime (V_D = 0.1 V). Both TFTs feature a channel width of 500 μm, Mo electrodes, and a gate dielectric of SiO_2 (100 nm). (c) Comparison of the field-effect mobility between the 10 and 100 μm channel lengths.

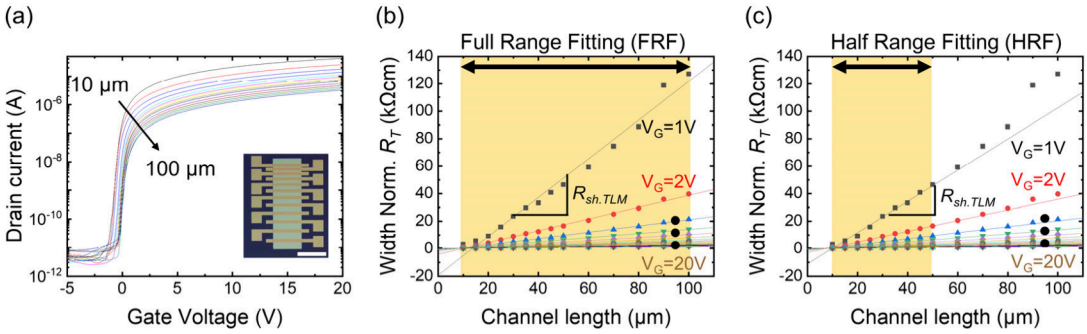

Figure 2. (a) Transfer curves of TLM devices with a constant width of 500 μm, measured over a channel length range from 10 to 100 μm at 0.1 V of drain voltage. The scale bar indicates a length of 0.5 mm. (b) The width-normalized total resistance vs channel length characteristics based on full-range fitting and (c) half-range fitting.

contacts, highlighting the significant impact of R_C on mobility extraction, especially as the channel length decreases.[34]

To analyze devices with high R_C, the transmission line method (TLM) is typically used to evaluate the sheet resistance (R_{sh}) of the device, thereby extracting the effective channel length (L_{eff}) and R_C.[35−37] This helps to correct mobility calculation errors due to channel length modulation. However, as the device dimensions decrease, the reliability of extracting the $R_{Sh.TLM}$ value using TLM becomes increasingly uncertain. This observation is mainly caused by the gate-dependent change in total resistance (R_T) in relation to R_{Ch} and R_C. R_T includes R_{Ch}, R_C, and interelectrode metal resistance (R_M). Typically, the R_M value is negligible, allowing us to simplify the expression as $R_T = R_{Ch} + R_C$. In long-channel TFTs, changes in R_{Ch} due to the gate exhibit a significant impact on R_T. However, in short-channel TFTs, as the channel length decreases, the influence of R_{Ch} diminishes and the influence of R_C increases. This results in a different gate dependence of R_T compared to long-channel TFTs because the R_{Ch} and R_C gate dependencies are not the same. Consequently, TLM extracts R_{Sh} based on the gate dependence of R_T, which can lead to variations in the extracted $R_{Sh.TLM}$ value depending on the channel length. This reduces the reliability of the evaluation.

Therefore, here, we used the gated van der Pauw (gVDP) method to overcome the limitations of incorrect mobility calculations caused by increased R_C, as the channel length decreases. The gVDP method allows for the extraction of the intrinsic $R_{Sh.gVDP}$ value of the thin film without being affected by R_C.[38−41] Unlike TLM, it does not lose reliability with changing channel lengths and can clearly distinguish between R_{Ch} and R_C even in short channels. This demonstrates that $R_{Sh.gVDP}$ directly extracted from gVDP can compensate for the model errors and overcome the issue of mobility over-estimation.

2. RESULTS AND DISCUSSION

2.1. Nonideal I_D−V_G Characteristics. A common approach for characterizing the electrical performance of oxide thin-film transistors is to employ the classic MOSFET model. This model can be expressed as follows in the linear mode where $|V_D| < |V_G - V_T|$.

$$I_D = \mu_{FE} C_{OX} \frac{W}{L}[(V_G - V_T)V_D - \frac{V_D^2}{2}] \quad (1)$$

where I_D is the TFT drain current; W and L denote the channel width and length, respectively; μ_{FE} is the field-effect mobility; C_{OX} stands for gate insulator capacitance per unit area; V_G and V_T are the gate voltage and threshold voltage, respectively; and V_D is the potential at the drain contact.

To investigate the impact of R_C on the electrical characteristics of a-IGZO TFTs, we examined the correlation between I_D and V_G depending on the channel length. The ideal I_D−V_G relationship, as described by eq 1, demonstrates a linear increase in I_D with increasing V_G in the linear mode. However, when R_C, particularly resulting from the Schottky contact between the metal electrode and active channel, becomes significant, the ideal I_D−V_G relationship may no longer be valid. Figure 1a illustrates the I_D−V_G characteristics of a long-channel a-IGZO TFT with a channel length of 100 μm, showing a linear increase in I_D with increasing V_G in the turn-on region. On the other hand, Figure 1b demonstrates the I_D−

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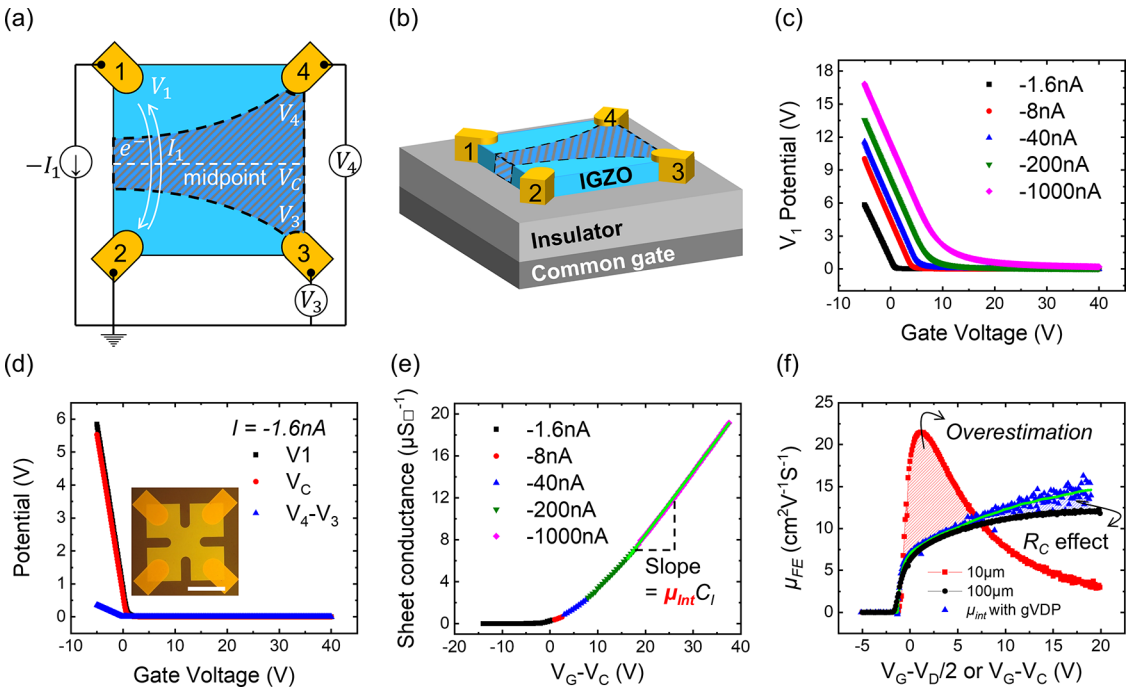

Figure 3. (a) A top view schematic of a van der Pauw (VDP) device with a square-shaped IGZO thin film semiconductor and (b) a 3D side view schematic of a gVDP device configuration fabricated on a common gate and insulator. (c) Potential V_1 values as a function of gate voltage driven at different currents. (d) Potentials of V_1, V_C, and $V_4 - V_3$, with V_C and $V_4 - V_3$ unaffected by R_C corresponding to $V_D/2$ and V_D, respectively. The scale bar indicates a length of 0.5 mm. (e) Extracted sheet conductivity driven by different currents. The fitted line is included in the curve, where the slope represents intrinsic mobility. (f) Comparison of μ_{FE} for 10 and 100 μm channel lengths and μ_{Int} of the IGZO channel extracted using the gVDP method.

V_G behavior of a short-channel a-IGZO TFT with a channel length of 10 μm, where the R_C effect is apparent, leading to a nonlinear increase in I_D with increasing V_G. This is because, when the R_C/R_T ratio increases with the reduced channel length, the gate dependence of the R_C affects the gate dependence of the R_T, and the potential drop due to R_C can no longer be ignored. Consequently, shorter channel lengths are more vulnerable to R_C, which can lead to a nonideal I_D−V_G relationship and can cause peak-shaped μ_{FE}, as shown in Figure 1c, leading to inaccurate device evaluation. Therefore, to improve the evaluation error due to the decrease in channel length, TLM and gVDP analysis methods were compared, and the precise $R_{Sh.gVDP}$ extraction of the gVDP analysis method effectively corrected the error.

2.2. Transmission Line Method. The TLM is typically used to characterize TFTs, as illustrated in Figure 2a. It is used to extract μ_{FE} by taking the derivative of eq 1 with respect to V_G. The μ_{FE} is defined as show in eq 2:[42]

$$\mu_{FE} = \frac{G_m}{C_{OX}V_D}\frac{L}{W} \tag{2}$$

G_m is the transconductance; $G_m = \partial I_D/\partial V_G$ and is determined from the transfer curve of the TFTs. To extract precise μ_{FE} using the TLM method, it is important to consider the channel length deviation (ΔL) due to depletion regions at the metal/semiconductor interface, as well as the physical difference between the actual and designed channel lengths, and the field effect mobility is given by $\mu_{FE} = \frac{G_m}{C_{OX}V_D}\frac{L-\Delta L}{W}$. Therefore, it is important to distinguish between R_{Ch} and R_C and accurately evaluate $L_{eff} = L - \Delta L$ using the following equations:

$$\begin{aligned} R_T &= V_D/I_D \\ &= R_{Ch} + R_C \\ &= R_{Sh}\frac{L-\Delta L}{W} + R_C \\ \therefore wR_T &= R_{Sh}(L-\Delta L) + wR_C \end{aligned} \tag{3}$$

Equation 3 expresses the linear relationship between R_T and L at a given V_G. Typically, the TLM measures the R_T values along the channel length, and we can obtain the slope of a linear line from the R_T vs L characteristic, which corresponds to the $R_{Sh.TLM}$. It allows for the extraction of ΔL and wR_C from the intersection of the lines, as shown in Figure 2b and 2c. However, the issue is that the slope of the line can be varied with fitting ranges if TFTs have a high R_C. For example, we compared two different cases. Figure 2b includes the full-range fitting (FRF) characteristic of TFTs with varying channel lengths, ranging from 10 to 100 μm. From the data, we can evaluate the $R_{Sh.TLM}$ value. In contrast, Figure 2c presents the half-range fitting (HRF) within the 10 to 50 μm range. Figure 2b and 2c show that the extracted $R_{Sh.TLM}$ values exhibit variation based on the chosen fitting range. The result arises from two main factors. First, the slope of R_T vs L measured in TLM reflects the average gate dependence of R_T, thus not accurately representing the exact values of individual components. Second, as the channel length decreases, the proportion of R_C in the R_T increases, distorting the gate dependence of the R_T trend. Consequently, the choice of fitting range significantly affects the extracted $R_{Sh.TLM}$ in TLM, thereby exerting a profound influence on the device evaluation. Therefore, since the $R_{Sh.TLM}$ value extracted by TLM based on the gate dependence of R_T varies with the channel length, it is

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not suitable for short-channel TFTs that require a precise R_{Sh} value. To overcome this issue, we used the van der Pauw method, which can extract the intrinsic $R_{Sh.VDP}$ value of the channel without being affected by R_C. The cases of electrodes with different contact resistances are included in Figure S1.

2.3. Gated van der Pauw Method. A gated van der Pauw (gVDP) method would address the issue in short-channel TFTs by accurately extracting the intrinsic $R_{Sh.VDP}$ value of the a-IGZO channel accurately. Figure 3a presents the basic topology of the VDP device including a square-shaped thin-film pattern with four corner contacts. The contact size should be significantly smaller than the square thin-film pattern, and using a cloverleaf-shaped thin-film pattern relaxes the contact size requirement and simplifies device fabrication. The electrical measurement setup for the VDP device involves applying a specific current I_1 as a source to contact 1, ground contact 2, and measure the potentials of V_3 and V_4 at contacts 3 and 4, respectively. The current density decreases when current flows in the film, creating an equipotential region. The potential at the midpoint of a channel between contacts 3 and 4 can be expressed as $V_C = \frac{|V_3 + V_4|}{2}$. This gives the potential of the middle region of the channel, which is equipotential to the V_C extracted through contacts 3 and 4. Figure 3b shows a gVDP device with an additional gate contact. This gate contact allows the changes in potential within the channel to be monitored, enabling a determination of the on/off state of the transistor based on the gate voltage. This information can be used to extract the sheet conductance ($\sigma_{sh.gVDP} = 1/R_{sh.gVDP}$).

The determination of $\sigma_{sh.gVDP}$ can be achieved by using the geometric characteristics of the VDP device. First, the current I_1 at contact 1 is used to calculate $R_{12} = \frac{|V_3 + V_4|}{I_1}$. Here, R_{12} is the resistance associated with a current flowing between contacts 1 and 2 and represents the intrinsic resistance inside the thin film where there is no voltage drop due to the contact resistance. Similarly, R_{21} can be calculated for a current flowing from contact 2 to contact 1. This process is repeated for a total of eight resistance values (R_{23}, R_{24}, R_{41}, ...). Subsequently, the average resistance value ($\bar{R}$, measured at each current in Figure S2) can be derived. This averaging process minimizes the impact of variables caused by the device's geometrical alignment error and the amorphous nature of the IGZO thin film, enhancing data precision. For the extraction of σ_{sh} at midpoint, the carrier concentration n is given by[40,42−45]

$$n = C_{OX}(V_G - V_T - V_X)/q, \quad \sigma = nq\mu \tag{4}$$

where V_X is the potential at distance X. Considering the potential at the center of the channel, for a typical MOSFET model, the midpoint of the channel is $\frac{V_D}{2}$ when $V_X = 0$ V, while for a gVDP, it is $V_C = \frac{V_3 + V_4}{2}$. On the other hand, the MOSFET model is simplified by the drift current equation:

$$I_D = W\sigma E_X, \quad E_X = (V_D - V_S)/L \tag{5}$$

Therefore, using eqs 4 and 5 to compute the sheet conductance for the typical MOSFET model and gVDP methods, the results are as follows:

$$\sigma_{sh} = \mu_{FE} C_{OX} \left| V_G - V_T - \frac{V_D}{2} \right| = \frac{I_D}{V_D} \frac{L}{W} \tag{6}$$

$$\sigma_{sh.gVDP} = \mu_{Int} C_{OX} |V_G - V_T - V_C| = \frac{I_1}{|V_4 - V_3|} \frac{\ln(2)}{\pi} \tag{7}$$

Here, μ_{FE} represents the field-effect mobility with R_C as current flows through the electrodes, while μ_{Int} represents the intrinsic mobility of the channel itself, which is not affected by R_C.

The reason is that in a gVDP configuration, there is no current flowing through the electrodes between contacts 3 and 4. Equations 6 and 7 exhibit variations based on whether there is a potential drop due to R_C or not. The potential at the midpoint of the channel is $\frac{V_D}{2}$ for a typical MOSFET model and $V_C = \frac{V_3 + V_4}{2}$ for gVDP. In the MOSFET model, the current I_D flows due to the applied V_D, while in gVDP, the potential V_4 −- V_3 is formed by the injected I_1. Moreover, $\frac{L}{W}$ is defined as $\frac{\ln(2)}{\pi}$ for the VDP geometry.[41] Therefore, the geometric design of gVDP devices simplifies the calculation of sheet conductance, $\sigma_{sh.gVDP} = \frac{\ln 2}{\pi \bar{R}}$.

Figure 3c shows the potential V_1 for each current in gVDP. The IGZO channel is n-type. Therefore, applying a negative current to contact 1 for electron carrier injection results in the generation of a V_1 potential. This phenomenon leads to two distinct operating regions depending on V_G. In the case of a positive V_G, V_1 converges to a very small value and decreases, whereas for a negative V_G, V_1 exhibits a linear relationship with V_G, both sharing the same slope. This behavior indicates the resistance characteristics during the on/off switching of current flow in transistor operation, corresponding to the linear and saturation modes. Figure 3d shows the potentials of V_1, V_C, and $V_4 - V_3$ at $I = -1.6$ nA. As previously discussed, in contrast to the typical values associated with a MOSFET model, $V_4 - V_3$ represents the potential difference between contacts 3 and 4, similar to V_D, while V_C denotes the potential at the midpoint of the channel, equivalent to $\frac{V_D}{2}$. As with V_1, the extrinsic potential, V_C, and $V_4 - V_3$ have the same on/off behavior mode depending on the current, which provides all the factors for extracting the intrinsic $\sigma_{sh.gVDP}$. Figure 3e presents the $\sigma_{sh.gVDP}$ values measured for different currents in the $V_G - V_C$ range from −5 to 40 V. The combined $\sigma_{sh.gVDP}$ values at different currents exhibit a linear relationship with $V_G - V_C$, showing that the characteristic is independent of R_C. To confirm this, we performed additional tests using gVDP devices with various electrodes, as shown in Figure S3a. The sheet conductance characteristics of gVDP devices were additionally investigated, depending on the sizes of the devices, in order to verify the size effect (Figure S3b). The reason is due to the amorphous nature of the IGZO channel or misalignment issue between the IGZO and electrode. The results demonstrate that the electrical characteristics of all devices remain consistent regardless of the size, attributed to a symmetric cloverleaf pattern that stabilizes voltage drops in the channel between the selected electrodes. According to eq 7, V_T can be determined by linear regression, simplifying the extraction of intrinsic mobility μ_{Int} = 14.69 cm^2 V^{-1} s^{-1} from the slope. Here, μ_{Int} is an intrinsic mobility of the channel itself, unaffected by R_C, thus indicating the maximum mobility achievable by the active channel material. Figure 3f compares the μ_{FE} values of 10 and 100 μm channel lengths and the μ_{Int} value of the IGZO channel. For the 10 μm channel length, the mobility extracted at a low V_G exceeds the actual maximum μ_{Int}

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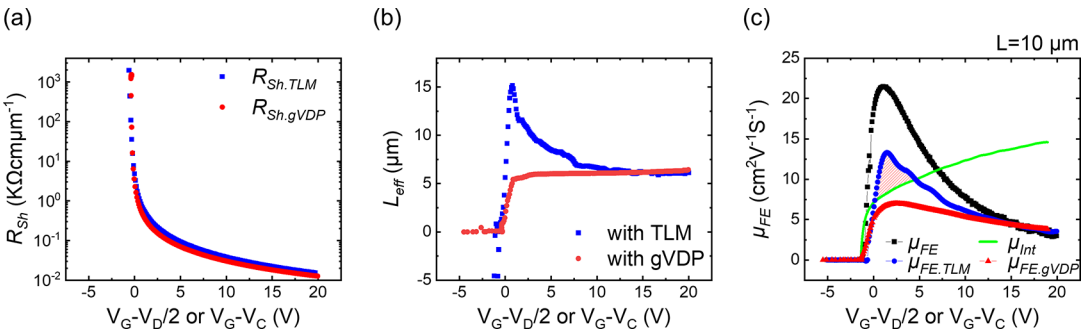

Figure 4. (a) Sheet resistance values extracted from the full-range fitting of TLM and gVDP devices. (b) ΔL and wR_C values obtained from $R_{Sh.TLM}$ and $R_{Sh.gVDP}$ data. (c) μ_{FE} extraction based on R_{Sh} extracted from TLM and gVDP.

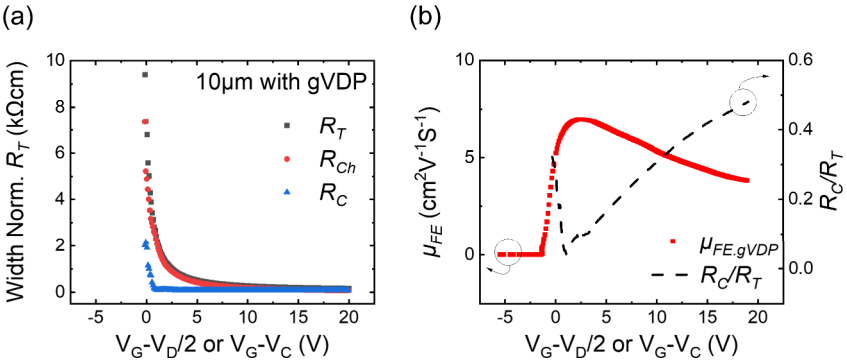

Figure 5. (a) Decoupled R_{Ch} and R_C values from the R_C value for a 10 μm channel length using $R_{Sh.gVDP}$ extracted from the gVDP analysis. (b) Comparison of the μ_{FE} curve based on gVDP and the change in the R_C/R_T ratio. All devices have a width of 500 μm.

value of the channel. This overestimation is attributed to the non-negligible R_C, leading to artificial errors. In contrast, for the 100 μm channel length, the influence of R_C is minimal, resulting in mobility values nearly identical to μ_{Int}. However, as V_G increases, the effect of R_C causes mobility saturation. The R_C effect refers to the phenomenon where, at high V_G, the relative magnitude of the R_C becomes more significant than the R_{Ch}, causing the gate dependence of R_T to be dominated by the gate dependence of the R_C. Therefore, the gVDP method enables the straightforward and accurate extraction of the intrinsic conductivity $\sigma_{sh.gVDP} = 1/R_{sh.gVDP}$ of the IGZO channel by mitigating variability associated with the W/L ratio through the geometric characteristics of the VDP device, independent of R_C. Additionally, μ_{Int} can be easily extracted from $\sigma_{sh.gVDP}$ and compared with μ_{FE} to evaluate the impact of R_C, making it a powerful analytical method. This approach also demonstrates greater precision than the TLM ($R_{sh.TLM}$) fitting method, which relies on channel length, showing significant potential for evaluating short-channel TFTs.

2.4. TLM vs gVDP Method. To compensate for the mobility overestimation of the MOSFET model, which occurs when analyzing transistors with significant R_C, the R_{Sh} value was precisely extracted through TLM and gVDP methods to correct the artificial errors. Figure 4a compares the values of $R_{Sh.TLM}$ and $R_{Sh.gVDP}$ based on $V_G - V_C$ or $V_G - V_D/2$. In the case of TLM, as the channel length shortens, the gate dependence of R_C affects the gate dependence of R_T, distorting the R_{Sh} value. Therefore, we used $R_{Sh.TLM}$ extracted from a wide range of fitting ranges to exclude the influence of R_C as much as possible. When $V_G - V_C$ or $V_G - V_D/2$ is a high potential, both $R_{Sh.TLM}$ and $R_{Sh.gVDP}$ show similar values, while at low potential, there is a slope difference between $R_{Sh.TLM}$ and $R_{Sh.gVDP}$. This is because the TLM device analysis using the MOSFET model calculates based on the average potential, $V_D/2$, without considering the potential drop that occurs to overcome the contact barrier at low potential, resulting in artificial error. On the other hand, gVDP considers the potential within the channel that does not generate R_C due to V_C, thus extracting an undistorted $R_{Sh.gVDP}$ value. Figure 4b shows the $L_{eff} = L - \Delta L$ extracted using $R_{Sh.TLM}$ and $R_{Sh.gVDP}$, respectively, to compensate for the peak mobility overestimation that occurred in the 10 μm channel length IGZO TFT. At high potential, since the contact barrier has already been overcome, the L_{eff} extracted by $R_{Sh.TLM}$ and $R_{Sh.gVDP}$ converge to similar values. However, at low potential, ΔL of the TLM is negatively extracted, resulting in an L_{eff} of 15.69 μm. This indicates that L_{eff} is excessively calculated to compensate for the underestimation of $R_{Sh.TLM}$ due to the contact barrier. Therefore, $R_{Sh.TLM}$ extracted based on the gate dependence trend of R_T, which varies with the fitting range, not only is an average value that cannot represent each individual device but also fails to completely exclude the effect of R_C, making it unsuitable for application to individual devices. In contrast, for gVDP, potential changes due to the contact barrier do not affect the extraction of $R_{Sh.gVDP}$. Therefore, even at a low potential, the L_{eff} does not exceed the actual value of 10 μm, allowing for the extraction of a precise L_{eff}. This method can be applied to short channels without resulting in artificial errors. Figure 4c shows the result of compensating for mobility overestimation through $R_{Sh.TLM}$ and $R_{Sh.gVDP}$ to verify how effectively the parameter calculation of the MOSFET model can be compensated. In the case of pristine, influenced by R_c, the potential applied to the channel is incorrectly calculated in the low potential region overcoming

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the contact barrier, resulting in parameter errors and ultimately causing a peak shape mobility overestimation of μ_{FE} = 21.48 $cm^2 V^{-1} s^{-1}$. The mobility of $\mu_{FE.TLM}$ and $\mu_{FE.gVDP}$, which was compensated by calculating channel length modulation through the inherent R_{Sh} of the channel material, corrects excessively high mobility, but $\mu_{FE.TLM}$ still shows unreliable results due to extremely high L_{eff} at low potential. On the other hand, $\mu_{FE.gVDP}$ effectively corrected the mobility error based on the $R_{Sh.gVDP}$ value extracted without being influenced by R_c. This demonstrates the effectiveness of using R_{Sh} to correct for artificial errors in the MOSFET model. In particular, it shows that the precise $R_{Sh.gVDP}$ value obtained from gVDP improves mobility errors even at low potentials, thereby demonstrating superior analytical reliability in short-channel TFTs compared to TLM.

Figures 5a and 5b show the decoupled R_{Ch} and R_C values by using the precisely extracted $R_{Sh.gVDP}$ value from the gVDP method, presenting the impact of R_C. Figure 5a demonstrates the different gate dependencies of R_{Ch} and R_C. R_{Ch} decreases exponentially over the entire potential range, while R_C shows a sharp decrease in the $V_G - V_C$ or $V_G - V_D/2$ range below the transition point of 0.75 V, where the slope of the curves abruptly changes. After this point, the current injection is significantly enhanced, inferring that the injection mechanism turns into the field emission from thermionic emission. When $V_G - V_C$ or $V_G - V_D/2$ exceeds the transition point of 0.75 V, the R_T values are significantly reduced by an order of magnitude from the initial value of 10 kΩ·cm. After this point, the field emission (tunneling) mechanism becomes dominant instead of thermionic emission, keeping the R_C low and constant. The R_C value was measured to be 0.08 kΩ·cm, similar to the reported value.[46,47] The results for the other electrodes are shown in Figure S4. Figure 5b presents the μ_{FE} and R_C/R_T ratio curves depending on the applied voltages. As we applied a bias higher than the transition point (0.75 V), the R_C/R_T value increases since the R_C remains constant, while the R_{Ch} continues to decrease exponentially, as shown in Figure 5a. Therefore, it causes a gradual mobility reduction. This observation aligns with Figure 3f, where mobility saturates or decreases due to the R_C effect.

The drain current model of MOSFETs relies on calculated scalar values and does not consider R_C, resulting in artificial errors in measured currents, such as nonlinear $I_D - V_G$, unlike the ideally predicted linear $I_D - V_G$ currents. To correct these errors, the distorted parameter R_{Sh}, intrinsic to the channel and influenced by R_C, was replaced with measured values instead of calculated ones. The comparison of the gVDP and TLM methods highlights the importance of precise R_{Sh} for the accurate characterization of IGZO short-channel TFTs. However, TLM extracts $R_{Sh.TLM}$ based on the gate dependence of the R_T trend, which hinders precise $R_{Sh.TLM}$ extraction due to drain potential modulation caused by potential drops within the channel. Additionally, it was found that $R_{Sh.TLM}$, which varies depending on the fitting range, becomes less reliable as the channel length decreases. This issue is not limited to IGZO TFTs but extends to various transistors, including 2D and organic semiconductors, where achieving ohmic contact at the source and drain contacts is challenging. In contrast, the gVDP method extracts $R_{Sh.gVDP}$ based on the measured potential within the channel, V_C, unaffected by R_C, effectively improving mobility overestimation. Moreover, this method not only extracts the μ_{Int} of the channel but also improves the accuracy of the mobility estimation. It can analyze the μ_{Int} of the channel to compare with μ_{FE} and precisely distinguish between R_{Ch} and R_C. This allows verification of the mechanism changes in R_C that cause mobility extraction errors before the mechanism transition and observation of the R_C effect that causes mobility saturation after the transition.

3. CONCLUSION

This study applied the gVDP method to address the issue of mobility overestimation caused by R_C in IGZO TFTs with short channel lengths. The gVDP method allows for the precise extraction of $R_{Sh.gVDP}$ of the channel without being affected by R_C, effectively correcting the artificial errors of the MOSFET model. The conventional MOSFET model does not account for R_C, leading to mobility overestimation in TFTs with short channel lengths or high R_C. The commonly used TLM is unreliable for extracting $R_{Sh.TLM}$ in short channels because it is not completely free from R_C effects. In contrast, the gVDP method can accurately extract $R_{Sh.gVDP}$ without potential drops caused by contact barriers, enabling precise extraction of the μ_{Int}. This method provides a standard for determining whether mobility is over- or underestimated compared to μ_{FE} and allows for easy identification of mobility saturation due to R_C. Additionally, the gVDP method distinguishes between R_{Ch} and R_C, enabling the identification of changes in the R_C mechanism and proving itself as a unique analytical tool for accurately evaluating R_C. The gVDP method would be valuable, not only for evaluating the electrical performance of IGZO TFTs but also for various transistors facing similar issues, including 2D and organic semiconductors. By improving the reliability of the mobility evaluation, the gVDP method can significantly contribute to the advancement of future electronics and semiconductor technology. This enhanced analytical capability is expected to have a substantial impact on the performance evaluation and future development of high-performance electronic devices.

4. EXPERIMENTAL SECTION/METHODS

4.1. Fabrication. All devices were fabricated on highly doped p-type Si wafers with thermally grown 100 nm SiO_2 as the gate oxide for the typical Si gate. SC-1 and SC-2 cleaning processes were performed before fabrication. The a-IGZO thin film was deposited by RF sputtering (located at the 3D Convergence Center of Inha University), followed by annealing in air at 350 °C in a furnace for 1 h. Patterning was then performed by wet etching using a HCl solution. The source/drain metals were deposited by using DC sputtering and patterned by using the lift-off method. All processes were performed simultaneously on TLM and gVDP devices in the same sample, and they shared the same Si common gate.

4.2. Characterization. All devices were measured in ambient air using a Keithley 4200 parameter analyzer connected to a probe station. Depending on the split length, the TLM device measurements were conducted with constant ground/voltage/voltage sweep conditions for the source/drain/gate contacts. The van der Pauw measurement method was carried out to analyze the gVDP device. Contact 1 provided the current; contact 2 was grounded; contacts 3 and 4 were used for voltage measurements on the gate contact while applying a voltage sweep. Considering the n-type behavior of a-IGZO, a negative current was applied to inject the majority carriers. Transistor measurements were first performed on contacts 1 and 2 and the gate contact to avoid stress due to the large potential difference between V_1 and the gate. The maximum current was set to 1 μA based on the measured currents. Eight measurements were taken, alternating between the four contact points, to reduce the variability due to defects. The sheet conductance for each current was aggregated to obtain the saturated sheet conductance as a function of

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the gate voltage. The intrinsic mobility was extracted according to eq 7.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.4c14405.

Analysis of TLM devices with various electrodes; van der Pauw average resistance extraction data; gVDP sheet conductance for different electrodes and sizes; contact resistance extraction using gVDP (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jeong-Hwan Lee − *Department of Materials Science and Engineering, Inha University, Incheon 22212, Republic of Korea; Program in Semiconductor Convergence and 3D Convergence Center, Inha University, Incheon 22212, Republic of Korea;* orcid.org/0000-0002-2276-4912; Email: jeong-hwan.lee@inha.ac.kr

Rino Choi − *Department of Materials Science and Engineering, Inha University, Incheon 22212, Republic of Korea; Program in Semiconductor Convergence and 3D Convergence Center, Inha University, Incheon 22212, Republic of Korea*; Email: Rino.choi@inha.ac.kr

Jiyoung Kim − *Department of Materials Science and Engineering, The University of Texas at Dallas, Richardson, Texas 75080, United States;* orcid.org/0000-0003-2781-5149; Email: jiyoung.kim@utdallas.edu

Authors

Woo-Seok Lee − *Department of Materials Science and Engineering, Inha University, Incheon 22212, Republic of Korea; Program in Semiconductor Convergence, Inha University, Incheon 22212, Republic of Korea;* orcid.org/0009-0008-4039-3159

Jaeho Lee − *Department of Materials Science and Engineering, Inha University, Incheon 22212, Republic of Korea; Program in Semiconductor Convergence, Inha University, Incheon 22212, Republic of Korea*

Amarja Katware − *Department of Materials Science and Engineering, Inha University, Incheon 22212, Republic of Korea; Program in Semiconductor Convergence, Inha University, Incheon 22212, Republic of Korea*

Noh-Hwal Park − *3D Convergence Center, Inha University, Incheon 22212, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.4c14405

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Notes

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■ REFERENCES

(1) Nomura, K.; Ohta, H.; Takagi, A.; Kamiya, T.; Hirano, M.; Hosono, H. Room-temperature fabrication of transparent flexible thin-film transistors using amorphous oxide semiconductors. *Nature* **2004**, *432* (7016), 488−492.

(2) Rha, S. H.; Jung, J.; Jung, Y.; Chung, Y. J.; Kim, U. K.; Hwang, E. S.; Park, B. K.; Park, T. J.; Choi, J.-H.; Hwang, C. S. Performance Variation According to Device Structure and the Source/Drain Metal Electrode of a-IGZO TFTs. *IEEE Trans. Electron Devices* **2012**, *59* (12), 3357−3363.

(3) Fortunato, E.; Barquinha, P.; Martins, R. Oxide semiconductor thin-film transistors: a review of recent advances. *Adv. Mater.* **2012**, *24* (22), 2945−2986.

(4) Kwon, J. Y.; Jeong, J. K. Recent progress in high performance and reliable n-type transition metal oxide-based thin film transistors. *Semicond. Sci. Technol.* **2015**, *30* (2), 024002.

(5) Kim, M.; Cho, S.-Y.; Shin, Y.-S.; Seok, Y.-C.; Kim, H.-W.; Yoon, J.-Y.; Choi, R.; Lee, J.-H. Improving Electrical Stability of a-InGaZnO Thin-Film Transistors with Thermally Deposited Self-Assembled Monolayers. *Electron. Mater. Lett.* **2020**, *16* (5), 451−456.

(6) Lee, J.-H.; Ke, T.-H.; Genoe, J.; Heremans, P.; Rolin, C. Overlapping-Gate Organic Light-Emitting Transistors. *Adv. Electron. Mater.* **2019**, *5* (1), 1800437.

(7) Choi, S.; Kim, B.; Jeong, J. K.; Song, Y. H. A Novel Structure for Improving Erase Performance of Vertical Channel NAND Flash With an Indium-Gallium-Zinc-Oxide Channel. *IEEE Trans. Electron Devices* **2019**, *66* (11), 4739−4744.

(8) Hara, Y.; Kikuchi, T.; Kitagawa, H.; Morinaga, J.; Ohgami, H.; Imai, H.; Daitoh, T.; Matsuo, T. IGZO-TFT technology for large-screen 8K display. *J. Soc. Inf. Dispersion* **2018**, *26* (3), 169−177.

(9) Mo, F.; Tagawa, Y.; Jin, C.; Ahn, M.; Saraya, T.; Hiramoto, T.; Kobayashi, M. Low-Voltage Operating Ferroelectric FET with Ultrathin IGZO Channel for High-Density Memory Application. *IEEE J. Electron Devices Soc.* **2020**, *8*, 717−723.

(10) Jang, H. J.; Lee, J. Y.; Kwak, J.; Lee, D.; Park, J.-H.; Lee, B.; Noh, Y. Y. Progress of display performances: AR, VR, QLED, OLED, and TFT. *J. Inf. Dispersion* **2019**, *20* (1), 1−8.

(11) Kim, T.; Choi, C. H.; Hur, J. S.; Ha, D.; Kuh, B. J.; Kim, Y.; Cho, M. H.; Kim, S.; Jeong, J. K. Progress, Challenges, and Opportunities in Oxide Semiconductor Devices: A Key Building Block for Applications Ranging from Display Backplanes to 3D Integrated Semiconductor Chips. *Adv. Mater.* **2023**, *35* (43), No. e2204663.

(12) Kamiya, T.; Hosono, H. Material characteristics and applications of transparent amorphous oxide semiconductors. *NPG Asia Mater.* **2010**, *2* (1), 15−22.

(13) Kwon, J.-Y.; Lee, D.-J.; Kim, K.-B. Review paper: Transparent amorphous oxide semiconductor thin film transistor. *Electron. Mater. Lett.* **2011**, *7* (1), 1−11.

(14) Heo, K. J.; Kim, H. S.; Lee, J. Y.; Kim, S. J. Filamentary Resistive Switching and Capacitance-Voltage Characteristics of the a-IGZO/TiO(2) Memory. *Sci. Rep.* **2020**, *10* (1), 9276.

(15) Besleaga, C.; Radu, R.; Balescu, L.-M.; Stancu, V.; Costas, A.; Dumitru, V.; Stan, G.; Pintilie, L. Ferroelectric Field Effect Transistors Based on PZT and IGZO. *IEEE J. Electron Devices Soc.* **2019**, *7*, 268−275.

(16) Mo, F.; Saraya, T.; Hiramoto, T.; Kobayashi, M. Reliability characteristics of metal/ferroelectric-HfO2/IGZO/metal capacitor for non-volatile memory application. *Appl. Phys. Express* **2020**, *13* (7), 074005.

(17) Kim, J.-S.; Jang, J.-H.; Kim, Y.-D.; Byun, J.-W.; Han, K.; Park, J.-S.; Choi, B.-D. Dynamic Logic Circuits Using a-IGZO TFTs. *IEEE Trans. Electron Devices* **2017**, *64* (10), 4123−4130.
(18) Tian, G.; Chen, J.; Yan, G.; Li, L.; Song, Z.; Yang, S.; Zhang, Z.; Xu, G.; Yin, H.; Yang, S.; et al. Highly Reliable Logic-in-Memory by Bidirectional Built-in Electric-Field-Modulated Multistate IGZO/AFE Nonvolatile Memory. *ACS Appl. Electron. Mater.* **2023**, *5* (2), 1041−1049.
(19) Lee, S. Y.; Kim, D. H.; Chong, E.; Jeon, Y. W.; Kim, D. H. Effect of channel thickness on density of states in amorphous InGaZnO thin film transistor. *Appl. Phys. Lett.* **2011**, *98* (12), 122105.
(20) Münzenrieder, N.; Salvatore, G. A.; Petti, L.; Zysset, C.; Büthe, L.; Vogt, C.; Cantarella, G.; Tröster, G. Contact resistance and overlapping capacitance in flexible sub-micron long oxide thin-film transistors for above 100 MHz operation. *Appl. Phys. Lett.* **2014**, *105* (26), 263504.
(21) Knobelspies, S.; Takabayashi, A.; Daus, A.; Cantarella, G.; Münzenrieder, N.; Tröster, G. Improvement of contact resistance in flexible a-IGZO thin-film transistors by CF4/O2 plasma treatment. *Solid-State Electron.* **2018**, *150*, 23−27.
(22) Yang, C.-I.; Tsai, C.-M.; Yu, M.-C.; Zhang, S.; Chang, T.-C.; Liao, P.-Y.; Chen, L.-H.; Chen, B.-W.; Chou, W.-C.; Chen, G.-F.; et al. Drain-Induced-Barrier-Lowing-Like Effect Induced by Oxygen-Vacancy in Scaling-Down via-Contact Type Amorphous InGaZnO Thin-Film Transistors. *IEEE J. Electron Devices Soc.* **2018**, *6*, 685−690.
(23) Huang, X. D.; Song, J. Q.; Lai, P. T. Improved Performance of Scaled-Down α-InGaZnO Thin-Film Transistor by Ar Plasma Treatment. *IEEE Electron Device Lett.* **2016**, *37* (12), 1574−1577.
(24) Lee, D.-H.; Kwon, Y.-H.; Seong, N.-J.; Choi, K.-J.; Kim, G.; Yoon, S.-M. Analysis on Contact Resistance and Effective Channel Length of Thin Film Transistors Using Composition-Modified In-Ga-Zn-O Active Channels Prepared with Atomic Layer Deposition and Various Electrode Materials. *ACS Appl. Electron. Mater.* **2022**, *4* (12), 6215−6228.
(25) Cho, E. N.; Kang, J. H.; Yun, I. Contact resistance dependent scaling-down behavior of amorphous InGaZnO thin-film transistors. *Curr. Appl. Phys.* **2011**, *11* (4), 1015−1019.
(26) Samanta, S.; Han, K.; Sun, C.; Wang, C.; Kumar, A.; Thean, A. V.-Y.; Gong, X. Amorphous InGaZnO Thin-Film Transistors With Sub-10-nm Channel Thickness and Ultrascaled Channel Length. *IEEE Trans. Electron Devices* **2021**, *68* (3), 1050−1056.
(27) Xie, Q.; Xu, J.; Taur, Y. Review and Critique of Analytic Models of MOSFET Short-Channel Effects in Subthreshold. *IEEE Trans. Electron Devices* **2012**, *59* (6), 1569−1579.
(28) Nasr, J. R.; Schulman, D. S.; Sebastian, A.; Horn, M. W.; Das, S. Mobility Deception in Nanoscale Transistors: An Untold Contact Story. *Adv. Mater.* **2019**, *31* (2), No. e1806020.
(29) Wong, H.-S.; White, M. H.; Krutsick, T. J.; Booth, R. V. Modeling of transconductance degradation and extraction of threshold voltage in thin oxide MOSFET's. *Solid-State Electron.* **1987**, *30* (9), 953−968.
(30) Uemura, T.; Rolin, C.; Ke, T. H.; Fesenko, P.; Genoe, J.; Heremans, P.; Takeya, J. On the Extraction of Charge Carrier Mobility in High-Mobility Organic Transistors. *Adv. Mater.* **2016**, *28* (1), 151−155.
(31) Liu, C.; Li, G.; Di Pietro, R.; Huang, J.; Noh, Y.-Y.; Liu, X.; Minari, T. Device Physics of Contact Issues for the Overestimation and Underestimation of Carrier Mobility in Field-Effect Transistors. *Phys. Rev. Appl.* **2017**, *8* (3), 034020.
(32) Bittle, E. G.; Basham, J. I.; Jackson, T. N.; Jurchescu, O. D.; Gundlach, D. J. Mobility overestimation due to gated contacts in organic field-effect transistors. *Nat. Commun.* **2016**, *7*, 10908.
(33) Pei, K.; Chen, M.; Zhou, Z.; Li, H.; Chan, P. K. L. Overestimation of Carrier Mobility in Organic Thin Film Transistors Due to Unaccounted Fringe Currents. *ACS Appl. Electron. Mater.* **2019**, *1* (3), 379−388.
(34) Okachi, T. Mobility overestimation due to minority carrier injection and trapping in organic field-effect transistors. *Org. Electron.* **2018**, *57*, 34−44.
(35) Kim, W.-S.; Moon, Y.-K.; Kim, K.-T.; Lee, J.-H.; Ahn, B.-d.; Park, J.-W. An investigation of contact resistance between metal electrodes and amorphous gallium-indium-zinc oxide (a-GIZO) thin-film transistors. *Thin Solid Films* **2010**, *518* (22), 6357−6360.
(36) Lee, J. K.; An, S.; Lee, S.-Y. Self-Aligned Top-Gate IGZO TFT With Stepped Structure for Suppressing Short Channel Effect. *IEEE Electron Device Lett.* **2023**, *44* (11), 1845−1848.
(37) On, N.; Kim, B. K.; Lee, S.; Kim, E. H.; Lim, J. H.; Jeong, J. K. Hot Carrier Effect in Self-Aligned In-Ga-Zn-O Thin-Film Transistors With Short Channel Length. *IEEE Trans. Electron Devices* **2020**, *67* (12), 5544−5551.
(38) Jiang, S.; Qian, J.; Wang, Q.; Duan, Y.; Guo, J.; Zhang, B.; Sun, H.; Wang, X.; Liu, C.; Shi, Y.; et al. Probing Coulomb Interactions on Charge Transport in Few-Layer Organic Crystalline Semiconductors by the Gated van der Pauw Method. *Adv. Electron. Mater.* **2020**, *6* (7), 2000136.
(39) Bonafè, F.; Decataldo, F.; Fraboni, B.; Cramer, T. Charge Carrier Mobility in Organic Mixed Ionic-Electronic Conductors by the Electrolyte-Gated van der Pauw Method. *Adv. Electron. Mater.* **2021**, *7* (10), 2100086.
(40) Chen, C.; Hu, S.; Chen, Z.; Shen, Y.; Liu, C.; Wu, Q.; Yang, T.; Liu, C. Generalized Gated Four-Probe Method for Intrinsic Mobility Extraction With Van Der Pauw Structure. *IEEE Electron Device Lett.* **2020**, *41* (2), 244−247.
(41) Rolin, C.; Kang, E.; Lee, J. H.; Borghs, G.; Heremans, P.; Genoe, J. Charge carrier mobility in thin films of organic semiconductors by the gated van der Pauw method. *Nat. Commun.* **2017**, *8*, 14975.
(42) Chen, C.; Yang, B. R.; Li, G.; Zhou, H.; Huang, B.; Wu, Q.; Zhan, R.; Noh, Y. Y.; Minari, T.; Zhang, S.; et al. Analysis of Ultrahigh Apparent Mobility in Oxide Field-Effect Transistors. *Adv. Sci. (Weinh)* **2019**, *6* (7), 1801189.
(43) Jeong, J.; Jun Lee, G.; Kim, J.; Moon Jeong, S.; Kim, J.-H. Analysis of temperature-dependent electrical characteristics in amorphous In-Ga-Zn-O thin-film transistors using gated-four-probe measurements. *J. Appl. Phys.* **2013**, *114* (9), 094502.
(44) Chen, C.-Y.; Kanicki, J. Gated-four-probe a-Si: H TFT structure: A new technique to measure the intrinsic performance of a-Si: H TFT. *IEEE Electron Device Lett.* **1997**, *18* (7), 340−342.
(45) van der Pauw, L. J. A method of measuring the resistivity and Hall coefficient on lamellae of arbitrary shape. *Philips Technol. Rev.* **1958**, *20*, 220−224.
(46) Ueoka, Y.; Ishikawa, Y.; Bermundo, J. P.; Yamazaki, H.; Urakawa, S.; Osada, Y.; Horita, M.; Uraoka, Y. Effect of Contact Material on Amorphous InGaZnO Thin-Film Transistor Characteristics. *Jpn. J. Appl. Phys.* **2014**, *53*, 03CC04.
(47) Hu, W.; Peterson, R. L. Molybdenum as a Contact Material in Zinc Tin Oxide Thin-Film Transistors. *Appl. Phys. Lett.* **2014**, *104* (19), 192105.

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